

Customer Churn Exploratory Data Analysis Report

1. Project Title

Customer Churn - Exploratory Data Analysis (EDA)

2. Objective

The main objective of this project is to perform Exploratory Data Analysis on the Telco Customer Churn dataset. The analysis focuses on understanding customer behavior, identifying factors affecting customer churn, and finding important patterns through visualization.

3. Dataset Description

The dataset used for this analysis is the Telco Customer Churn Dataset.

Dataset Details:

- Number of Records: 7043
- Number of Features: 21
- Target Variable: Churn

Important Features:

customerID, gender, SeniorCitizen, Partner, Dependents, tenure, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges, Churn.

4. Data Cleaning and Preprocessing

- Loaded dataset using Python Pandas.
- Checked dataset structure and feature information.
- Verified missing values.
- Converted TotalCharges into numerical format.
- Checked duplicate records.
- Analyzed numerical and categorical variables.
- Prepared dataset for visualization.

5. Exploratory Data Analysis

EDA was performed to understand customer churn patterns using statistical analysis and visualization.

6. Data Visualization and Analysis

6.1 Customer Churn Distribution

Chart Type: Count Plot

Description:

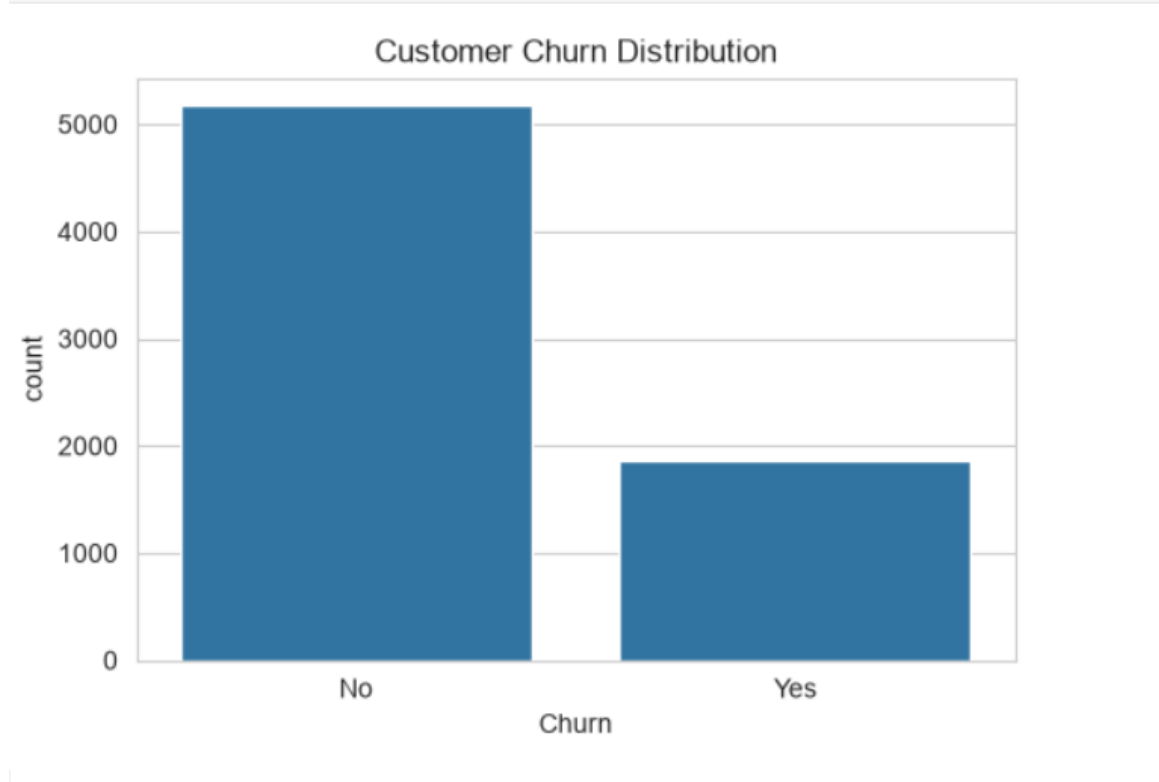
This visualization shows the distribution of customers based on their churn status.

Observation:

The graph represents customers who stayed with the service and customers who discontinued the service.

Insight:

It helps understand the overall distribution of churned and retained customers.



6.2 Churn Percentage Pie Chart

Chart Type: Pie Chart

Description:

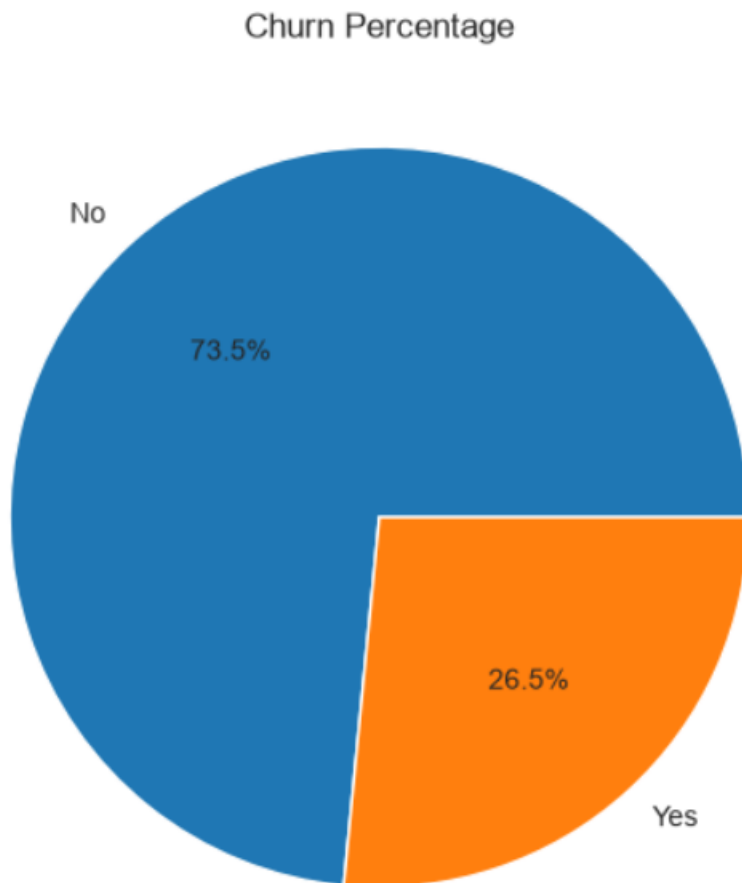
The pie chart represents percentage distribution of churned and retained customers.

Observation:

It provides a clear view of customer loss and retention percentage.

Insight:

This helps identify the overall churn rate.



6.3 Numerical Feature Histogram

Chart Type: Histogram

Description:

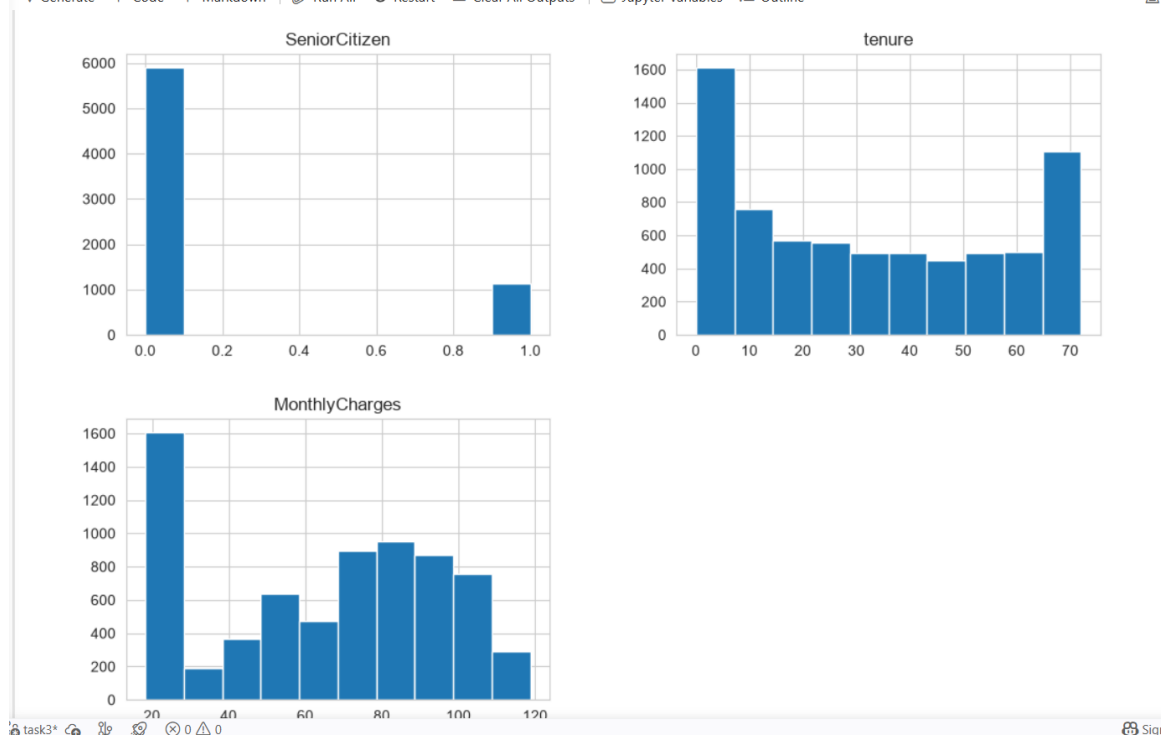
Histograms were used to analyze numerical features such as tenure, monthly charges and total charges.

Observation:

The distribution of numerical values helps understand customer behavior.

Insight:

Numerical feature patterns can influence customer churn.



6.4 Boxplot (Outlier Detection)

Chart Type: Box Plot

Description:

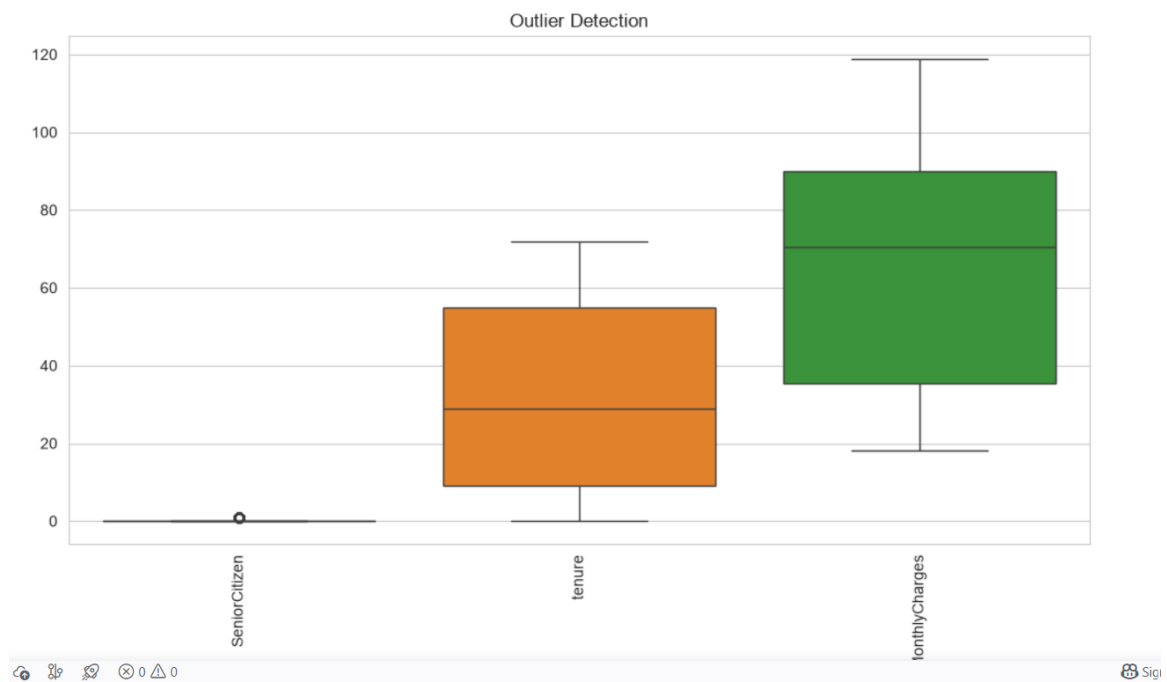
Boxplots were used to detect outliers in numerical features.

Observation:

The graph displays feature spread and unusual values.

Insight:

Outlier detection improves data quality and analysis accuracy.



6.5 Churn by Gender

Chart Type: Bar Plot

Description:

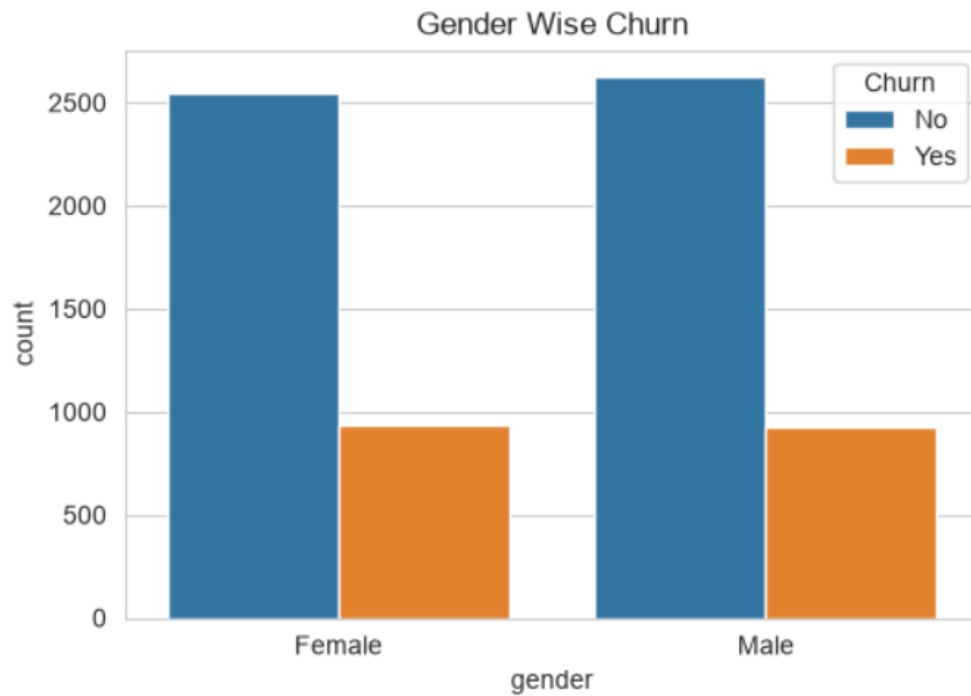
Customer churn was analyzed based on gender.

Observation:

The visualization compares churn behavior between male and female customers.

Insight:

It helps understand gender-based churn patterns.



6.6 Churn by Contract Category

Chart Type: Bar Plot

Description:

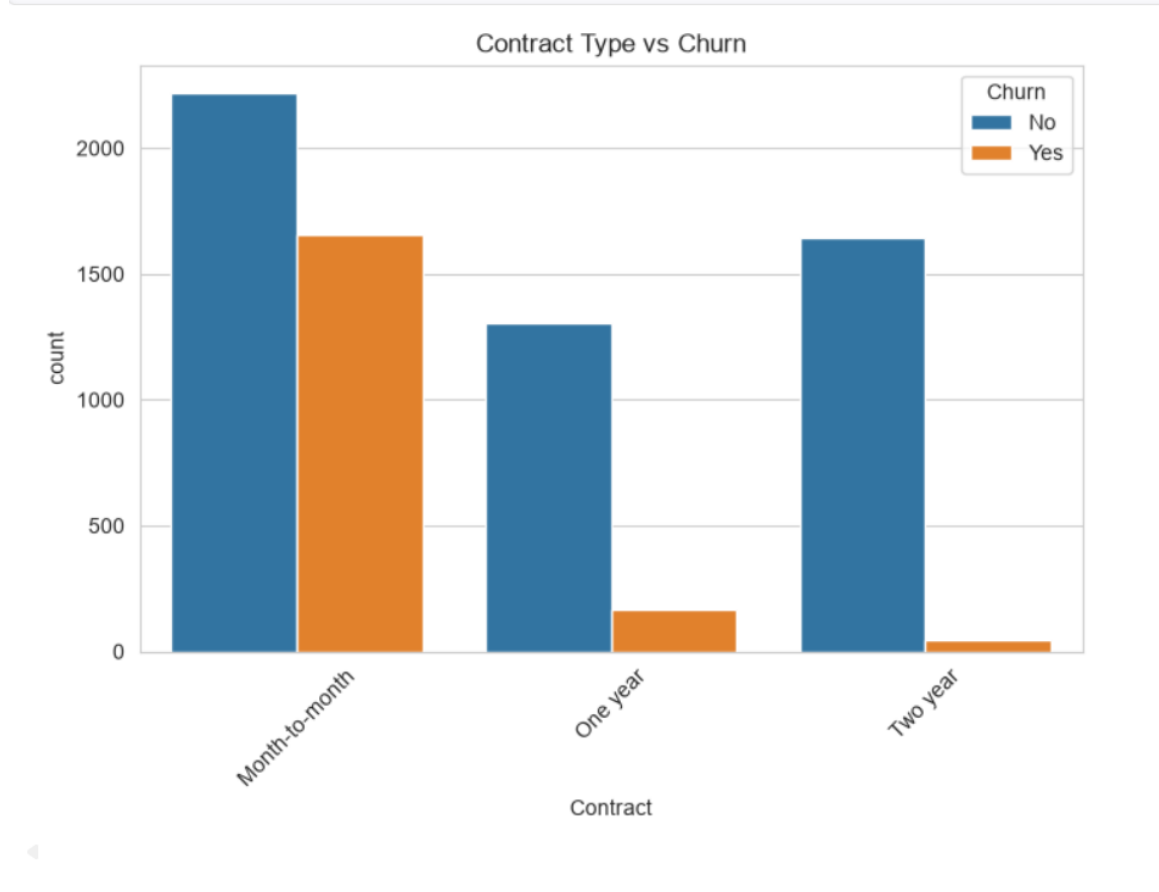
Customer churn was analyzed based on contract categories.

Observation:

Month-to-month, one year, and two year contracts were compared.

Insight:

Short-term contract customers generally show higher churn probability.



6.7 Correlation Heatmap

Chart Type: Heatmap

Description:

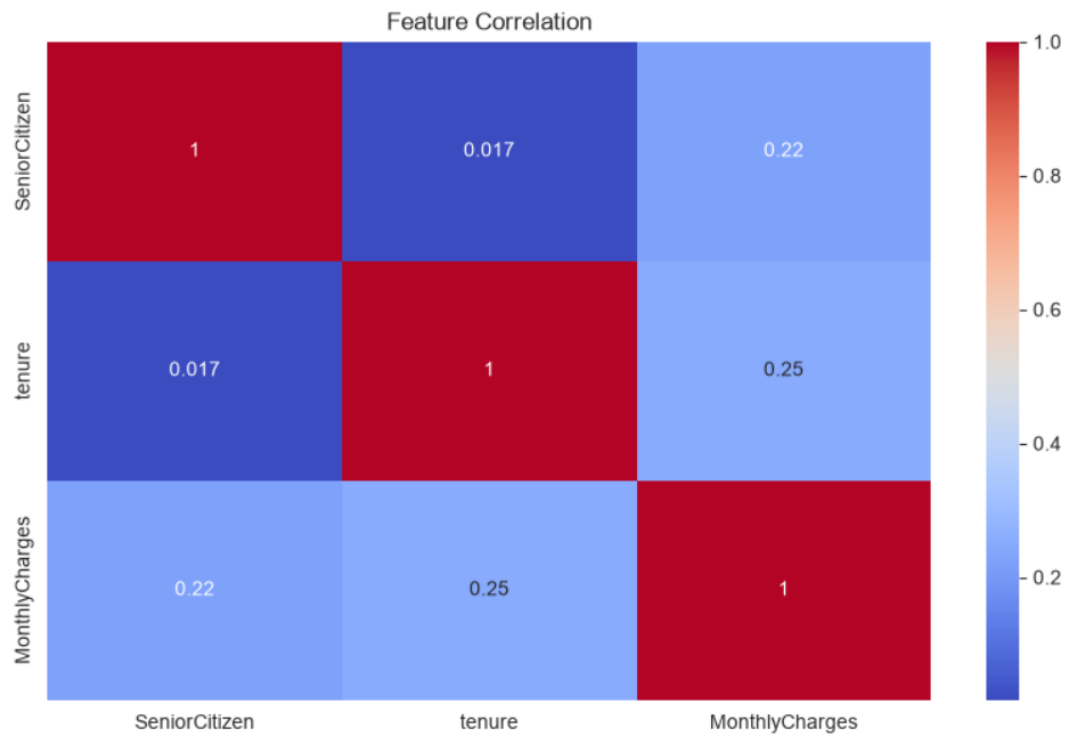
A correlation heatmap was created to identify relationships between numerical features.

Observation:

The heatmap shows relationships between variables.

Insight:

It helps identify important features affecting churn.



6.8 Scatter Plot Analysis

Chart Type: Scatter Plot

Description:

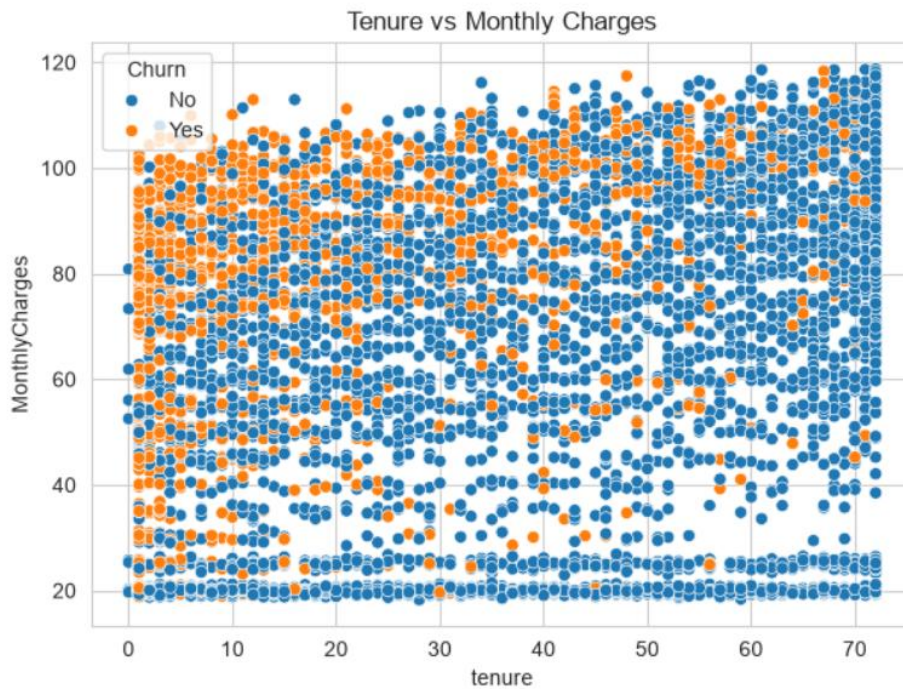
Scatter plot was used to analyze relationships between numerical features.

Observation:

It shows patterns between features such as tenure and charges.

Insight:

It helps understand trends between numerical variables.



7. Key Findings

- Contract type is an important factor influencing churn.
- Customers with shorter tenure have higher churn probability.
- Monthly charges affect customer retention.
- Long-term customers are more likely to continue services.
- Service features influence customer decisions.

8. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

- Jupyter Notebook
- Visual Studio Code

9. Conclusion

The Exploratory Data Analysis helped identify customer churn patterns and important factors affecting customer retention. The insights can be used to improve customer retention strategies.